

Logistic growth in small cohorts

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Abstract

We study how three small cohorts grow over five weeks and show that a logistic model fits the observed counts better than exponential growth. The fitted growth rate r is 0.9 per week for cohort A. Figure 1 shows the trend, Table 1 lists the raw counts, and Section 2 describes the setup. Earlier work [1] established the baseline method that we compare against.

1 Introduction

Population growth in a limited environment slows as it approaches the carrying capacity. The key relation is the logistic model, where the population P at time t follows $P(t) = K / (1 + A e^{(-r t)})$. Here K is the carrying capacity, r the growth rate, and A is a constant fixed by the initial value. Equation (1) on the next page states it formally.

Readers often lose their place when a figure is on another page. In this viewer, a pop-out card shows Figure 1 next to the sentence that mentions it, so you never have to scroll away. Hold the Option key and point at this paragraph to have it explained in plain words.

1.1 Motivation

Small cohorts are common in pilot studies, where noise is high and the number of observations is low. A model with few parameters that still captures saturation is therefore useful in practice, and the logistic curve is the simplest such model.

2 Setup

Three cohorts were observed over five weeks. The chart below summarises the trend for cohort A; the table lists the exact weekly counts for all cohorts.

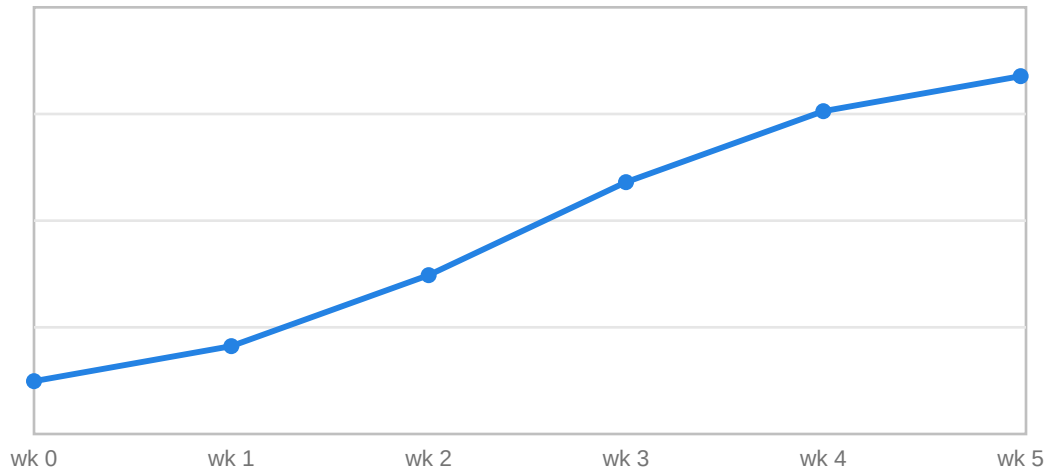


Figure 1: Population of cohort A over five weeks with the fitted logistic curve.

Table 1: Weekly counts per cohort.

Week	Cohort A	Cohort B	Cohort C
1	12	9	15
2	20	14	22
3	41	30	38
4	66	52	60
5	80	71	77

$$P(t) = K / (1 + A e^{(-r t)}) \quad (1)$$

Equation (1) is fitted by least squares on the weekly counts. The resulting growth rate r is 0.9 per week for cohort A, with a carrying capacity K of about 85.

3 Enrolment form

Full name:

Date of birth (DD/MM/YYYY):

Email:

Phone: 555-0134-9921 ID number: AB1234567

I agree to the terms: ☐

Cohort: ☐ A ☐ B

Country:

Signature: _____

Date: _____

References

[1] A. Author and B. Writer. A baseline method for cohort growth. Journal of Tests, 12(3):45-67, 2021.

[2] C. Person. Another reference. Proceedings of Things, 2020.